

Adaptive Dual Motion Model for Facial Micro-Expression Generation

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ABSTRACT

Facial micro-expression (ME) refers to a brief spontaneous facial movement that can reveal the genuine emotion of a person. The absence of data is a major problem for ME. Thankfully, generative deep neural network models can aid in producing desired samples. In this work, we proposed a deep learning based adaptive dual motion model (ADMM) for generating facial ME samples. A dual motion extraction (DME) module extracts robust motions from two modalities: original color images and edge-based grayscale images, with dual streams. Using edge-based grayscale images can help the method focus on learning subtle movements by eliminating the influences of noise and illumination variants. The motions extracted by the dual streams are fed into an adaptive motion fusion (AMF) module for combining the motions adaptively to generate the dense motion. Our method was trained on the CASME II, SMIC, and SAMM datasets. The evaluation and analysis of the results demonstrated the effectiveness of our method.

CCS CONCEPTS

• **Information systems** → Multimedia content creation; • **Computing methodologies** → Computer vision representations; • **Human-centered computing** → Human computer interaction.

KEYWORDS

facial micro-expression generation; adaptive dual motion model; edge-based motion extraction

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1 INTRODUCTION

Facial micro-expression (ME) stands for a brief spontaneous facial movement (less than a half second), which can unravel the genuine emotion of a person [25]. ME was discovered as a micro-momentary expression and a nonverbal leakage around the middle of the 20th

century [5, 7]. ME has a variety of applications, including national security, political psychology, medical care, and so on [22].

ME analysis includes ME spotting and recognition [1]. In recent years, ME analysis has moved from traditional machine learning methods using handcraft features to deep learning based methods with automatically learned features [1, 12, 14, 18, 21, 22, 29, 30]. Although there is a great success in ME analysis, limited available data still slow down the ME development [11]. Facial micro-expression generation (MEG) can help alleviate this problem by creating new data. Supervised MEG methods utilized action unit (AU) annotations to extract AU-related features, and generate new images with the condition of these AUs [24]. Unsupervised or self-supervised MEG methods demonstrated superior performance in the micro-expression grand challenge (MEGC) 2021 [6, 28]. The methods learn motion features from input videos themselves, and create new images based on the learned motion. However, the subtle ME movements can be easily affected by noise and illumination changes, resulting in an unsatisfactory motion extraction.

In this work, we proposed a deep learning based adaptive dual motion model (ADMM) for generating facial ME samples. A dual motion extraction (DME) module extracts robust motions from two modalities: original color images and edge-based grayscale images, with dual streams. Using edge-based grayscale images can help the method focus on learning subtle movements by eliminating the influences of noise and illumination variants. The motions extracted by the dual streams are fed into an adaptive motion fusion (AMF) module for combining the motions adaptively to generate the dense motion. Our method was trained on the CASME II, SMIC, and SAMM datasets. The evaluation and analysis of the results demonstrated the effectiveness of our method. The source code of our work is available online¹.

The remainder of the paper is structured as follows: Section 2 examines the pertinent literature. Section 3 describes the approach employed. Section 4 contains the datasets, implementation details, evaluation, analysis, and discussions. Section 5 closes the paper.

2 RELATED WORK

2.1 Facial Micro-Expression Analysis

The objective of ME spotting is to determine the duration in which MEs occur [1]. In contrast to previous work using optical flows as inputs [9, 20, 27], Yap *et al.* proposed a 3D convolutional neural network (3D-CNN) to learn motions using temporal oriented reference frames directly [26]. ME recognition refers to the categorization of MEs into emotion types [1]. Attention modules and expression-specific learnings were suggested and utilized in [21, 30], respectively. Even if ME analysis is on the rise, it is still a challenging task to synthesize new images to increase its sample size.

¹<https://github.com/xinqi-fan/adaptive-dual-motion-model>

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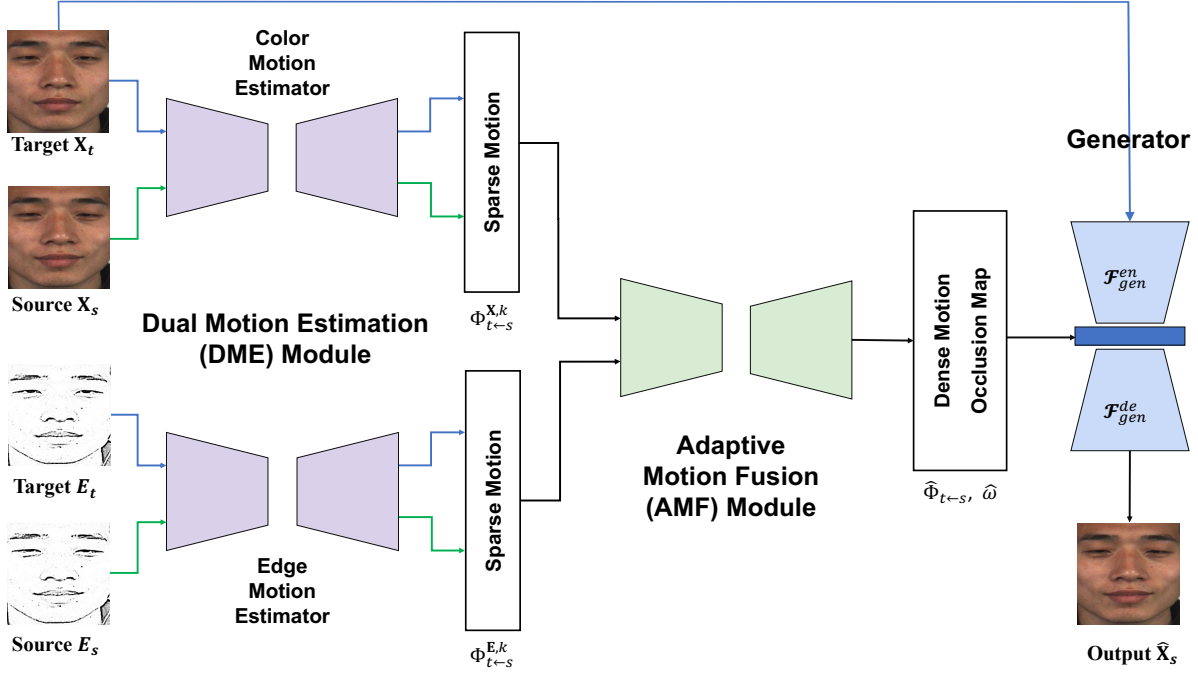


Figure 1: The pipeline of the ADMM during self-supervised training. The DME module extracts motions from original color images and edge-based grayscale images, respectively. The AMF module adaptively combines the two modalities' motions to form the dense motion. During testing, the target, source, and output frames should be replaced as Y , X_τ and $\hat{Y}_\tau$ with their corresponding edges at the time stamp τ .

2.2 Facial Micro-Expression Generation

MEG aims to generate new images based on the identity of the target image with the ME of the source images. There are two types of methods presented in MEGC 2021: feature based and motion based methods. The feature based method [24] retrieved facial AUs from the source sequences, and produced new images conditioned on the AUs of the source images. Zhang *et al.* [28] and Fan *et al.* [6] both used the motion based methods improved from the first order motion model (FOMM) [19], which retrieved facial motion in the source sequences, and retargeted it to the target image. Due to the strong localized and disentangled motion extraction capability of FOMM, this work is developed based on it but focuses on enhancing its performance to eliminate the effects of noise and illumination variations.

3 METHODOLOGY

Given a source facial ME sequence $\mathbf{X} = \{X_1, \dots, X_L\}$ and a static target facial image Y , the task of MEG is to synthesize a new facial sequence $\hat{\mathbf{Y}} = \{\hat{Y}_1, \dots, \hat{Y}_L\}$ with the face in Y and the ME in $\mathbf{X}$. In this work, we proposed a deep learning based ADMM for generating facial ME images in Figure 1. The DME module extracts robust motions from two modalities: original color images and edge-based grayscale images, with dual streams. Using edge-based grayscale images can help the method focus on learning subtle movements by eliminating the influences of noise and illumination variants. The

motions extracted by the two streams are fed into an AMF module for combining the motions adaptively to generate the dense motion.

3.1 Dual Motion Estimation Module

To help the model learn better subtle movements without the influence of noise and illumination variants, we proposed a DME module that extracts motions from both original color images and edge-based grayscale images. The DME module consists of two parts: a color motion estimator $\mathcal{F}_{motion}^X$ [19] to estimate motions from original color images, and an edge motion estimator $\mathcal{F}_{motion}^E$ to estimate motions from edge-based grayscale images.

The edge-based grayscale images are generated in advance by an extended difference of Gaussian (XDoG) operator [23]. Although the XDoG operator cannot be viewed as a standard edge detector like Canny [3], it can extract edge-like lines as well as preserve sufficient details of the original image features. These advantages of the XDoG operator make it suitable for the application of ME to help learn subtle movements. Given color images X_ϑ , their corresponding edge maps E_ϑ are generated by the XDoG operator G_E as

$$E_\vartheta = G_E(X_\vartheta) \in \mathbb{R}^{H \times W}, \quad \vartheta \in \{s, t\}, \quad (1)$$

where H is the height and W is the width of the input images.

By assuming that there exist abstract reference frames X_τ and E_τ for the original color inputs and the edge-based grayscale inputs, the estimators can generate sparse motions (affine transformation)

with respect to the reference frames as

$$\Phi_{g \leftarrow r}^\Gamma = \mathcal{F}_{motion}^\Gamma(\Gamma_g), \quad \Gamma \in \{X, E\}. \quad (2)$$

In addition, the motion estimators also output P keypoints for color inputs and Q keypoints for grayscale inputs. In practice, the motion from the color image stream $\Phi_{g \leftarrow r}^X$ consists of P local motions $\{\Phi_{g \leftarrow r}^{X,k} \in \mathbb{R}^{2 \times 3}\}_{k=1}^P$, while the motion from the edge-based grayscale image stream $\Phi_{g \leftarrow r}^E$ consists of Q local motions $\{\Phi_{g \leftarrow r}^{E,k} \in \mathbb{R}^{2 \times 3}\}_{k=1}^Q$ according to keypoints' locations.

After obtaining the motion from the reference frame to the source frame $\Phi_{s \leftarrow r}^{\Gamma,k} \in \mathbb{R}^{2 \times 3}$ and the motion from the reference frame to the target frame $\Phi_{t \leftarrow r}^{\Gamma,k} \in \mathbb{R}^{2 \times 3}$, we can combine them to obtain the desired sparse motion from the source to the target frame $\Phi_{t \leftarrow s}^{\Gamma,k}$ as

$$\Phi_{t \leftarrow s}^{\Gamma,k} = \Phi_{t \leftarrow r}^{\Gamma,k} \begin{bmatrix} \Phi_{s \leftarrow r}^{\Gamma,k} & \\ 0 & 1 \end{bmatrix}^{-1} \in \mathbb{R}^{2 \times 3}. \quad (3)$$

By replacing the Γ in $\Phi_{t \leftarrow s}^{\Gamma,k}$ with X and E in (3), we obtain the sparse motions for the original color image stream as $\Phi_{t \leftarrow s}^{X,k}$ and for the edge-based grayscale image stream as $\Phi_{t \leftarrow s}^{E,k}$.

3.2 Adaptive Motion Fusion Module

To effectively combine the motions estimated from the original color image stream and the edge-based grayscale image stream, we proposed to combine the motions from two streams adaptively using an AMF module, rather than simply adding or concatenating the motions. Similar to the dense motion estimation in [19], we utilized an encoder-decoder network to estimate weights for each motion, and combine them by a weighted combination.

To indicate the change around each location, P difference of Gaussian heatmaps $\{H_k\}_{k=1}^P$ between the source and target keypoints of the original color frames are computed as

$$H_k(z) = \exp\left(\frac{(\mathcal{F}_{motion}^{X,kp}(X_s) - z)^2}{\sigma}\right) - \exp\left(\frac{(\mathcal{F}_{motion}^{X,kp}(X_t) - z)^2}{\sigma}\right), \quad (4)$$

where $\mathcal{F}_{motion}^{X,kp}(X_g)$ indicates the keypoint outputs of $\mathcal{F}_{motion}^X(X_g)$, σ is the variance. Similarly, we compute Q heatmaps $\{H_k\}_{k=P+1}^K$ ($K = P + Q$) for the edge-based grayscale frames.

The target frame is also aligned using sparse motion $\Phi_{t \leftarrow s}^{\Gamma,k}$ to generate K aligned frames $\{\chi_t^k\}_{k=1}^K$. All aligned frames χ_t^k and heatmaps H_k are concatenated and passed to a fusion network $\mathcal{F}_{fusion}$ to generate the predicted weights ψ_k for each local motion obtained from the original color and edge-based grayscale streams as

$$\{\psi_k\}_{k=1}^K = \mathcal{F}_{fusion}(\{\chi_t^k\}_{k=1}^K, \{H_k\}_{k=1}^K). \quad (5)$$

Finally, we can obtain the dense motion $\hat{\Phi}_{t \leftarrow s}$ by fusing the sparse motion of the original color frames $\Phi_{t \leftarrow s}^{X,k}$ and the sparse motion of the edge-based grayscale frames $\Phi_{t \leftarrow s}^{E,k}$ through a linear combination using the predicted weights ψ_k as

$$\hat{\Phi}_{t \leftarrow s} = \sum_{k=1}^P \psi_k \Phi_{t \leftarrow s}^{X,k} + \sum_{k=P+1}^K \psi_k \Phi_{t \leftarrow s}^{E,k}. \quad (6)$$

3.3 Image Generator

Due to the fact that some portions of the target may not appear in the source or may be distorted during the deformation, the AMF module also generates an occlusion map $\hat{\omega}$ to indicate the location to be inpainted. Given the target frame X_t , we first pass it through an encoder $\mathcal{F}_{gen}^{en}(X_t)$ to obtain a compact embedding, and then apply the dense motion $\hat{\Phi}_{t \leftarrow s}$ on the embedding by a deformation operation $\mathcal{H}_w$. It is multiplied with the occlusion map $\hat{\omega}$ to obtain the final embedding as

$$Y = \hat{\omega} \odot \mathcal{H}_w(\mathcal{F}_{gen}^{en}(X_t), \hat{\Phi}_{t \leftarrow s}). \quad (7)$$

The predicted image is generated by a decoder as $\hat{X}_s = \mathcal{F}_{gen}^{de}(Y)$.

3.4 Training and Testing

Training. We used a feature reconstruction loss and two equivariance losses for the two modalities. The feature reconstruction loss is similar to the perception loss [10] as

$$\mathcal{L}_{rec}^{c,m}(\hat{X}_s, X_s) = \frac{1}{N_m^c} |VGG_m^c(\hat{X}_s) - VGG_m^c(X_s)|, \quad (8)$$

where m refers to the m th VGG-19 feature map and the c th input scale used for the VGG-19 network, N_m^c is the total number of pixels on the c th scale and the m th feature map.

The equivariance losses are applied on both original color images and edge-based grayscale images as

$$\mathcal{L}_{equiv}^\Gamma = |\mathcal{F}_{motion}^{\Gamma,kp}(\mathcal{T}_{rand}(\Gamma_s)) - \mathcal{T}_{rand}(\mathcal{F}_{motion}^{\Gamma,kp}(\Gamma_s))|, \quad (9)$$

where $\mathcal{T}_{rand}$ is a known random thin plate spline transformation.

Through a weighted summation, we can obtain the final loss as

$$\mathcal{L} = \sum_c \sum_m (\lambda_r^{c,m} \mathcal{L}_{rec}^{c,m}) + \lambda_{eq}^X \mathcal{L}_{equiv}^X + \lambda_{eq}^E \mathcal{L}_{equiv}^E, \quad (10)$$

where λ s are the weights.

Testing. Rather than directly extracting motion between source and target samples, relative motion is extracted from the source sequence itself. This relative motion is then applied to the target frame to generate the final deformed sequences. After the motion is extracted, some can also be amplified to make the ME more pronounced.

4 EXPERIMENT AND RESULT

4.1 Dataset

4.1.1 CASME II. CASME II [25] includes 26 participants and 255 ME sequences. All videos were captured at 200 frames per second (FPS) in 640×480 resolution. In particular, we resized the given cropped images to 256×256 pixels.

4.1.2 SMIC. SMIC-HS [15] consists of 164 spontaneous ME clips captured at 100 FPS with high-speed cameras from 16 individuals. SMIC-VIS data are captured at a slower rate. We utilized SMIC-HS and SMIC-VIS given cropped images by resizing them to 256×256 .

4.1.3 SAMM. In SAMM [4], there are 159 MEs. The onset and offset frames of the ME intervals are annotated. In particular, we used the OpenCV Viola Joins face detector to crop faces to 256×256 and aligned them with the first frame.

Table 1: Comparison with other methods. For each coder, the presented scores are normalized perfect scores, which are calculated by raw scores normalized by the perfect score ($9 \times 36 = 324$). Overall is the sum of the normalized perfect scores. Relative is the sum of normalized scores relative to the best method result from each coder.

Method	Coder 1	Coder 2	Coder 3	Overall	Relative
# 36 (Our)	0.61	0.56	0.44	1.61	2.46
# 46	0.73	0.55	0.58	1.86	2.82
# 63	0.74	0.58	0.66	1.98	3.00

4.2 Implementation Detail

Our model was developed using the PyTorch framework [17]. The DME and generator have been initialized from VoxCeleb pretraining [16], while the AMF has been initialized by He initialization [8]. They were optimized by an adaptive moment (Adam) optimizer with an initial learning rate of 2×10^{-6} and 2×10^{-4} for 50 epochs of training. A multistep learning rate scheduler was utilized to dynamically reduce the learning rate by ten at epochs 20 and 30. We used a high-performance computer system with 4 CPU cores, 1 NVIDIA V100 GPU card, and 32GB of memory to train our model. The target faces were cropped by the method in [2]. The number of keypoints were set as $P = Q = 10$, but they can be different.

4.3 Evaluation

For each database, there are 3 sequences, each containing positive, negative, and surprise expressions, selected as source samples. In terms of target samples [13], there are 4 samples that include an Asian female with glasses, an Asian male without glasses, a western male with glasses, and a western female without glasses, where the first two are selected from CAS(ME)³ [11]. In total, there are 36 videos, i.e., 4 templates $\times$ 3 emotions for each database. All videos are unified at 100 FPS, with a resolution of 256×256 .

According to the challenge guideline [13], three coders who hold facial action coding system (FACS) certificates from UK, Finland, and China were invited for evaluation based on AUs. Each labeling the results without interfering with the other. The results of the coder evaluation for the top three methods are available in Table 1, and more details can be found online².

4.4 Analysis and Discussion

The use of edge map. We tested the edge images with black and white to indicate the edges (Figure 2). We thought that the white-outlined edges should be preferred for emphasizing the key informational components, but its results indicated that the points are concentrated in the center, which is harmful to the performance. According to our hypothesis, too many 0 values in the inputs might make optimization difficult and result in a large number of neurons being dormant. Therefore, we chose edge images with black edges as the inputs to avoid optimization difficulties.

The visualization of adaptive weights. To validate the active role of the edge map inputs, we visualized some weights learned by the AMF module for the edge-based grayscale image stream

²<https://megc2022.github.io/challenge.html>

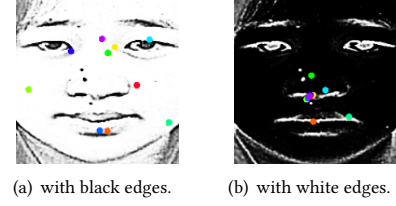


Figure 2: Visualization of learned keypoints.

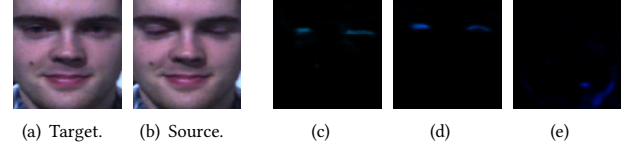


Figure 3: Visualization of weights for the edge-based grayscale stream motions.

motions in Figure 3 (c-e). The weights correspond to the motion extracted from the target and source color frames in Figure 3 (a-b). The visualization shows that the edge-based grayscale stream can contribute to some edge parts near eyes, eyebrows, outlines etc.

The motion generated around glasses. When the subject in the target samples wears glasses, we find that our technique is fairly robust (Figure 4). In the majority of the time, the movement of the eyelids will not have much of an impact on the glass.

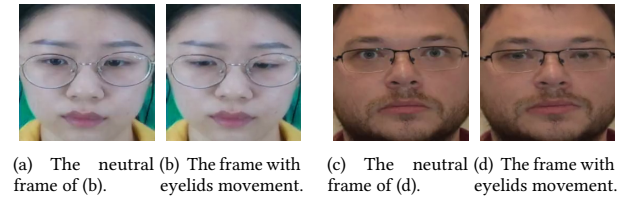


Figure 4: Visualization of eyelids movement with glasses.

5 CONCLUSION

In this work, we proposed a deep learning based ADMM method to generate facial ME samples. The DME module successfully extracted motion from two modalities: original color images and edge-based grayscale images, and the AMF module combined them adaptively as a dense motion. We trained the method on CASME II, SMIC, and SAMM datasets, and achieved satisfactory results. These results suggested the usefulness of the edge map against noise and illumination variants during motion extraction and generation. As it is critical to learn good keypoints for our method, future work will fully investigate the influence of keypoints and develop methods that are less prone to the errors in estimating these points.

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